

Food Delivery Management System

Decision Table & Decision Tree

Decision tables and decision trees are structured-analysis tools used to document conditional business logic in an unambiguous, testable form. This document models two decision points in the Food Delivery Management System: **Order Confirmation** (whether an order should be accepted) and **Delivery Fee Calculation** (what fee applies to a given order). Each is shown first as a decision table and then as the equivalent decision tree.

Tushar Kumar | 2400290120267

#	Artifact	Decision Modelled
1	Decision Table	Order Confirmation
2	Decision Tree	Order Confirmation
3	Decision Table	Delivery Fee Calculation
4	Decision Tree	Delivery Fee Calculation

1. Decision Table — Order Confirmation

Models whether an incoming order should be confirmed, based on four independent conditions. With 4 binary conditions there are 16 possible rules; each is shown below.

Rule	Restaurant Open	Items In Stock	Payment Successful	Address In Service Area	Action
R1	Y	Y	Y	Y	Confirm Order & Forward to Restaurant
R2	Y	Y	Y	N	Reject — Outside Delivery Area
R3	Y	Y	N	—	Reject — Payment Failed
R4	Y	N	—	—	Reject — Item(s) Out of Stock
R5	N	—	—	—	Reject — Restaurant Closed

Table 1: Order Confirmation Decision Table (— = don't care / short-circuited)

Reading the table

- Conditions are evaluated in priority order: Restaurant Open → Items In Stock → Payment Successful → Address In Service Area.
- A “—” means the condition is irrelevant once an earlier condition has already forced a rejection (short-circuit evaluation).
- Only Rule R1, where all four conditions are satisfied, results in the order being confirmed.

2. Decision Tree — Order Confirmation

The same logic as Table 1, expressed as a decision tree. Each internal node is a condition test; each leaf is the resulting action.

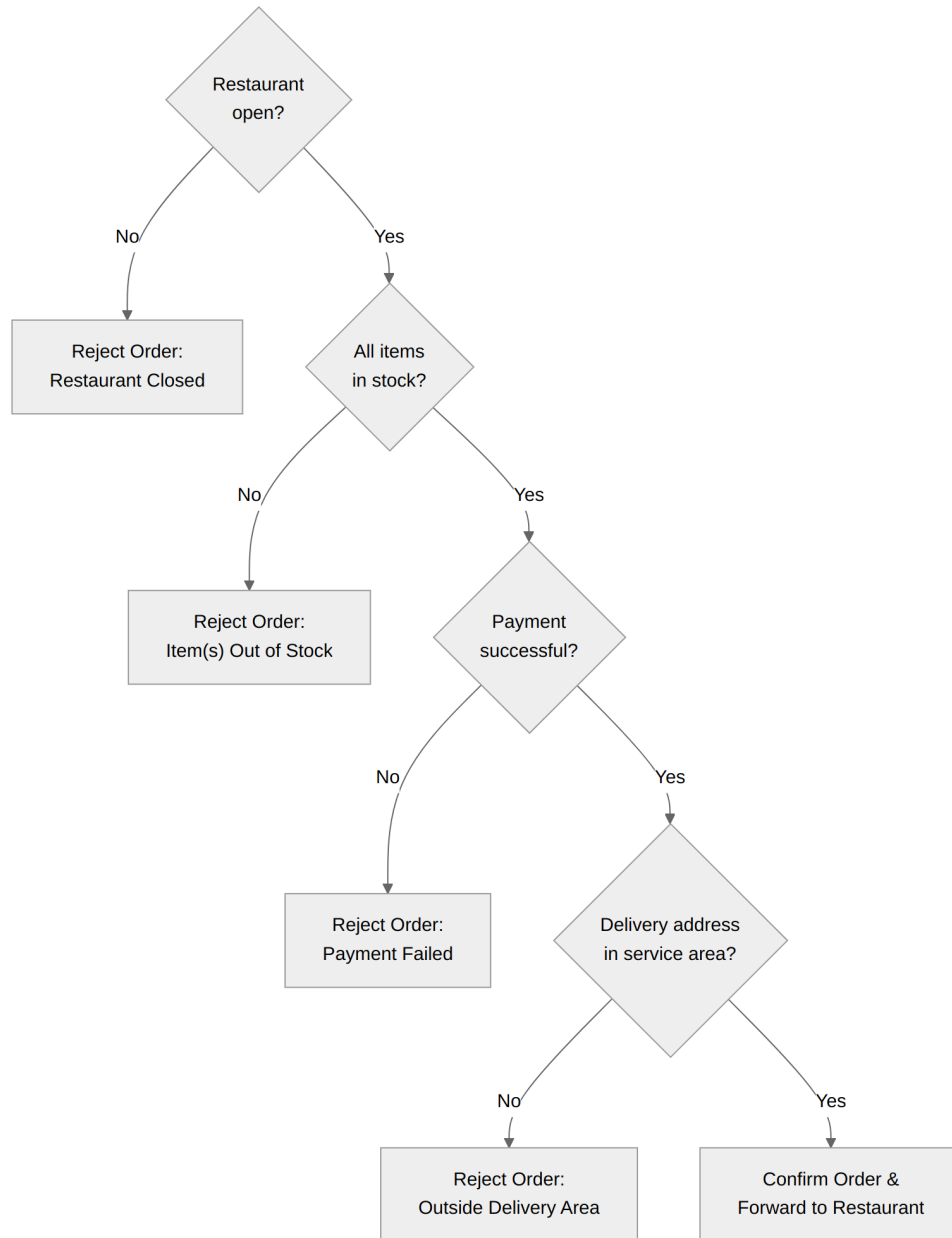


Figure 1: Decision Tree for Order Confirmation

3. Decision Table — Delivery Fee Calculation

Models the delivery fee charged to the customer, based on order value and delivery distance.

Rule	Order Value	Distance	Peak Hours	Delivery Fee
R1	$\geq$ Rs. 499	Any	–	FREE
R2	$<$ Rs. 499	≤ 3 km	–	Rs. 20
R3	$<$ Rs. 499	3–7 km	–	Rs. 40
R4	$<$ Rs. 499	> 7 km	No	Rs. 60
R5	$<$ Rs. 499	> 7 km	Yes	Rs. 60 + Surge

Table 2: Delivery Fee Decision Table

Reading the table

- Orders worth Rs. 499 or more always qualify for free delivery, regardless of distance.
- Below that threshold, the fee scales with distance in three bands: ≤ 3 km, 3–7 km, and beyond 7 km.
- A surge charge is added on top of the Rs. 60 base fee only for long-distance deliveries during peak hours.

4. Decision Tree — Delivery Fee Calculation

The same logic as Table 2, expressed as a decision tree, evaluated top-down for each incoming order.

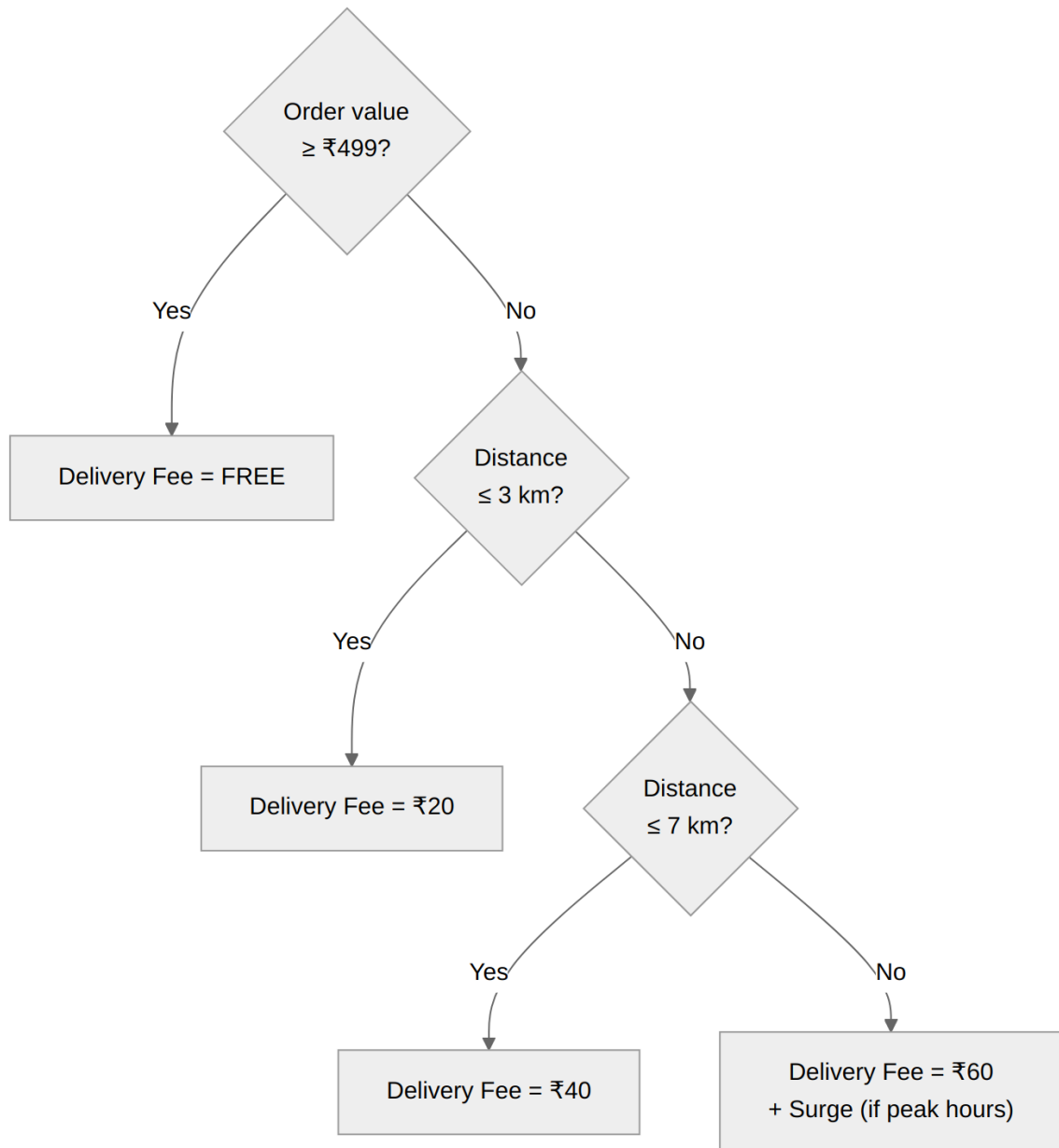


Figure 2: Decision Tree for Delivery Fee Calculation